



ALERTO
By NATHAN & FRIENDS

I. Rationale

Communities frequently experience disruptions involving electricity, water supply, internet connectivity, and ever-changing weather conditions, yet information regarding these incidents is often scattered across social media pages, unofficial announcements, and separate service provider platforms. This lack of centralized and accessible information can lead to confusion, delayed response, and reduced public preparedness during emergencies. With the rise of smart-city technologies, Geographic Information Systems (GIS), and data-driven platforms, this project aims to solve the challenges of fragmented infrastructure announcements, delayed public communication, and limited accessibility to emergency-related information within communities. By utilizing centralized digital technologies, environmental monitoring concepts, and GIS-based visualization, the proposed system seeks to enhance urban communication, infrastructure awareness, public preparedness, and community resilience through a more accessible and intelligent information platform.

Research on volunteered geographic information also highlights the importance of citizen participation in digital mapping and reporting systems, where residents contribute localized information that improves situational awareness and response coordination (Goodchild, 2007). To address these challenges, the proposed project, ALERTO, is a web-based information system that integrates GIS visualization, AI-assisted information retrieval, and community reporting into a centralized platform for infrastructure monitoring and emergency awareness. The system provides announcements related to power outages, water interruptions, internet service disruptions, and weather advisories while also allowing users to submit reports and access emergency contact information. By combining smart-city concepts with participatory GIS and AI-powered assistance, the platform promotes improved communication, accessibility, and community preparedness.

II. Whom is the solution for?

The proposed system is intended for local communities, government offices, utility providers, and emergency response agencies such as the Bureau of Fire Protection (BFP), Philippine National Police (PNP), hospitals, and MDRRMO offices. Residents often struggle with delayed, scattered, or inaccessible information regarding electricity outages, water supply interruptions, internet connectivity issues, weather advisories, and emergency situations, which can affect public safety, daily activities, and community preparedness. Likewise, local government units and service providers face challenges in efficiently disseminating announcements, monitoring incidents across different areas, and maintaining effective communication with the public. The proposed solution aims to address these issues by providing a centralized web-based platform with GIS map visualization, AI-assisted inquiry features, infrastructure announcement monitoring, emergency contact accessibility, and community reporting tools to improve communication, awareness, coordination, and emergency response within the community.

The proposed system is intended for the following users:

1. Residents and Citizens
2. Utility Providers
3. Local Government Units (LGUs)
4. Emergency Response Offices

III. Objectives

General Objective

To develop a web-based GIS-powered information system that centralizes infrastructure announcements, emergency information, and community reporting using AI-assisted information retrieval.

Specific Objectives

1. To provide interactive GIS-based visualization of affected areas and infrastructure announcements.
2. To centralize updates related to electricity, water supply, internet connectivity, and weather conditions.
3. To integrate an AI assistant capable of answering user queries based on official announcements and mapped locations, and to enhance the system with predictive analytics for essential services, including forecasts for potential power, water, and internet interruptions.
4. To provide accessible emergency contact information for different public offices and emergency services.
5. To allow users to submit infrastructure-related incident reports through the platform.

6. To improve communication efficiency between residents, utility providers, and local authorities.

IV. Software Solution Development Scope

Overview

ALERTO is a web-based information system designed to centralize infrastructure announcements, emergency information, and community-submitted reports through an interactive Geographic Information System (GIS) interface augmented by artificial intelligence. The system consolidates fragmented updates regarding electricity outages, water supply interruptions, internet connectivity issues, and weather advisories into a single accessible platform. This approach builds upon the established paradigm of volunteered geographic information, wherein citizens act as sensors by contributing localized data that enhances situational awareness and community response coordination (Goodchild, 2007). By integrating GIS visualization, AI-assisted query processing, and community reporting functionalities, ALERTO aims to bridge the communication gap between residents, utility providers, local government units (LGUs), and emergency response agencies. The platform enables users to view up-to-date infrastructure status on an interactive map, submit incident reports, query the system using natural language through an AI chatbot, and access emergency contact directories. This document outlines the complete technical scope of the software solution, including target platforms, intended users, system components, data flows, technology stack, security considerations, and AI-based features planned for development.

Target Platform

The system is developed as a progressive web application (PWA) , accessible through modern web browsers on both desktop and mobile devices. This platform choice was made to maximize accessibility without requiring users to install dedicated software. Research on web applications designed for marginalized communities has demonstrated that progressive web applications effectively lower barriers to access by enabling users to engage with digital services using devices they already own, without the friction of app store downloads or device storage limitations (Lindén Guinez, 2019). The web-based architecture ensures that residents with varying levels of technological literacy can access the system using devices they already own. The application is designed with responsive layouts to accommodate screen sizes ranging from mobile phones to desktop monitors.

Intended Users and Usage Environment

The system is designed for four primary user groups, each with distinct interaction patterns:

Residents and Citizens. Community members use the platform to view infrastructure announcements on the map, search for updates affecting their specific barangay or municipality, submit incident reports, and query the AI chatbot for summarized information. They access the system from personal devices in residential or mobile contexts. The inclusion of citizen reporting functionality is grounded in the principle that individuals on the ground possess the most immediate knowledge of local conditions, making them essential contributors to a comprehensive community awareness system (Goodchild, 2007).

Utility Providers. Personnel from electricity cooperatives, water districts, and internet service providers use the platform to post official announcements, monitor community-submitted reports that may indicate unreported outages, and communicate service restoration timelines. They access the system from office environments or field operations.

Local Government Units (LGUs). Municipal and barangay officials use the platform to monitor infrastructure status across their jurisdiction, coordinate responses to widespread disruptions, and publish official advisories. The GIS dashboard provides spatial awareness of incident clusters for resource allocation decisions.

Emergency Response Offices. Agencies including the Bureau of Fire Protection (BFP), Philippine National Police (PNP), hospitals, and Municipal Disaster Risk Reduction and Management Offices (MDRRMO) use the platform to access emergency contact directories, monitor incident locations, and coordinate with affected communities during emergencies.

The usage environment encompasses both routine daily monitoring and emergency scenarios where rapid access to reliable information is critical.

Core Modules, Pages, and System Components

The ALERTO system comprises the following core components:

- 1. Interactive GIS Map Module.** The primary interface for spatial visualization of all infrastructure reports and announcements. Features include color-coded map markers by incident category (electricity, water, internet, weather), clustering for high-density areas, municipality and barangay boundary overlays, click-to-detail popups showing report summaries, and a location search bar with autocomplete for navigating to specific areas. Users can hover over or select a municipality to view aggregated statistics and active reports.

2. Incident Reporting Module. A structured form interface allowing users to submit infrastructure-related incident reports. Inputs include incident category selection, textual description, location tagging via map pin drop or address search, optional photo upload, and reporter contact information. Upon submission, the report is geocoded, stored in the database, and immediately reflected on the GIS map. This module operationalizes the concept of citizens as sensors, transforming individual observations into actionable spatial data for community benefit (Goodchild, 2007).

3. AI Chatbot Module. A persistent floating chat interface accessible from any page. The chatbot processes natural language queries from users, retrieves relevant reports and announcements from the database, generates summarized responses, and provides predictive insights regarding potential future outages based on historical patterns stored in the system. This module employs a Retrieval-Augmented Generation (RAG) architecture, which research has shown achieves high accuracy for public service applications by combining precise information retrieval from structured databases with the conversational flexibility of large language models (Mubarak & Sutopo, 2025).

4. Announcements and Predictive Analytics Directory. A chronological feed of official announcements from utility providers and LGUs, filterable by category, municipality, and date range. This module displays AI-generated summaries, links to original announcements, and an integrated analytics section that provides AI-generated forecasts for possible infrastructure disruptions. Using historical report data, the system identifies patterns and predicts the likelihood and timing of future outages, supported by trend graphs and highlighted risk zones on the map.

5. Emergency Contacts Directory. A searchable and categorized directory of emergency contact numbers and office locations, including BFP, PNP, hospitals, MDRRMO, and local government offices. Entries are displayed both in list format and as pinned locations on the GIS map.

6. Administrative Panel. A secure backend interface for utility providers and LGU personnel to manage announcements, verify user-submitted reports, update incident statuses (reported, confirmed, resolved), and manage emergency contact entries.

Expected Inputs, Processes, and Outputs

Inputs. The system accepts the following data inputs: user-submitted incident reports containing category selection, textual descriptions, and geolocation data; official announcements from utility providers and LGUs; natural language queries submitted through the AI chatbot interface; location search queries entered in the map search bar; and administrative updates to report statuses and emergency contact information.

Processes. Core processing operations include: geocoding of address inputs to latitude and longitude coordinates for map placement; storage and indexing of reports in the database with spatial attributes; Retrieval-Augmented Generation (RAG) processing of chatbot queries against the report database and announcement repository an approach that has demonstrated precision rates as high as 98.5% and F1-scores of 98.8% in public information service chatbot applications (Mubarak & Sutopo, 2025); AI-powered summarization of lengthy announcement texts into concise summaries; predictive analysis using historical report data to identify temporal patterns and generate outage forecasts; and spatial querying to aggregate reports by municipality, barangay, or custom geographic boundaries.

Outputs. The system produces: an interactive GIS map with real-time report markers and boundary overlays; conversational responses from the AI chatbot containing retrieved and summarized information; a filterable feed of announcements with AI-generated result; visualizations on the map; status-updated report cards reflecting updates; and local emergency contact lists.

Planned Technologies and Technical Stack

The system is built using a modern web development stack divided into frontend, backend, database, and external service layers:

Frontend. React.js with functional components and hooks for state management. Leaflet.js integrated with OpenStreetMap tiles for GIS visualization. CSS with responsive design principles for cross-device compatibility. Axios for HTTP communication with the backend API.

Backend. Python Flask framework for RESTful API development. Flask-CORS for cross-origin resource sharing. Flask-SQLAlchemy for object-relational mapping and database management. Geopy library with Nominatim geocoding service for address-to-coordinate conversion.

Database. PostgreSQL with the PostGIS spatial extension for geospatial data storage and querying. The database stores user reports with latitude and longitude columns indexed for spatial queries, official announcements, emergency contact records, and municipality boundary data. For the initial development phase, SQLite may be used as a lightweight alternative with migration to PostgreSQL-PostGIS planned for production deployment, aligning with established practices for spatial database infrastructure that prioritizes ease of initial deployment while maintaining a clear upgrade path (Spatial DBStack, n.d.).

AI-Based Technologies. The system integrates three artificial intelligence features: (1) a Retrieval-Augmented Generation (RAG) chatbot that combines structured database queries with large language model response generation for answering user questions about infrastructure status, an architecture that has proven effective for municipal public information services by grounding responses in verified data rather than relying solely on generative model outputs (Mubarak & Sutopo, 2025); (2) an AI text summarization model that condenses lengthy

announcements and report descriptions into concise, scannable summaries; and (3) a predictive analytics component that uses time-series analysis of historical report data to forecast potential future outage events and identify high-risk periods or areas.

Security, Authentication, and Data Privacy Measures

The system implements role-based access control to differentiate between public users, utility providers, LGU personnel, and system administrators. Public users may view announcements, submit reports, and use the chatbot without authentication. Administrative functions, including announcement publishing and report status management, require secure login credentials. All API endpoints handling sensitive operations are protected with token-based authentication.

User-submitted report data is stored with minimal personally identifiable information; reporters may submit anonymously, and only optional contact details are retained for follow-up purposes. All data transmissions between the client and server are encrypted using HTTPS. Input validation and sanitization are implemented on all form submissions to prevent injection attacks. The database is configured with regular backup procedures and access logging.

In compliance with the Philippine Data Privacy Act (Republic Act No. 10173), a privacy notice is displayed to users, and data retention policies are defined. Geographic data is aggregated at the barangay level for public display, and precise coordinates of individual residences are not publicly exposed unless explicitly provided for public infrastructure reports.

Development Approach and Future Scalability

The project follows an iterative development methodology, with core functionalities prioritized for initial deployment. The minimum viable product includes the GIS map with report markers, the incident reporting form, the AI chatbot interface, and the emergency contacts directory. Subsequent iterations will add the predictive analytics dashboard, administrative panel, and enhanced filtering capabilities.

The system architecture is designed for scalability. The modular separation of frontend, backend API, and database layers allows each component to be scaled independently. The use of PostgreSQL with PostGIS ensures that as the volume of geospatial data grows with user adoption, queries remain performant through spatial indexing. Future expansion possibilities include integration with official utility provider APIs for automated outage detection, push notification support, and incorporation of additional data layers such as flood hazard maps, evacuation center locations, and real-time weather radar feeds to enhance the platform's role in community disaster preparedness and response.

V. Main Features

1. GIS Interactive Map

The GIS Interactive Map visualizes infrastructure and environmental reports on a geographic interface. It displays incidents such as power, water, and internet interruptions, along with weather advisories, in specific locations. Users can click areas to view detailed information like status and timestamps, helping them quickly understand affected zones and make informed decisions during disruptions.

2. AI Predictions and Announcements for Essentials Services

The system provides predictive analysis for essential utilities, including power interruptions, water interruptions, and internet stability, by analyzing historical data, usage trends, maintenance schedules, and network performance indicators such as latency and downtime. It uses Logistic Regression as its machine learning model to generate probability-based forecasts from observational and historical data to help anticipate possible service disruptions (John et al., 2022). It will also show announcements made by essential service providers to give a centralized platform for information dissemination.

3. AI Chatbot Feature (A.G.E.N.T)

The system includes a floating chatbot on the map view that users can open anytime. It allows users to ask questions in natural language about reported issues like power, water, or internet interruptions in a specific area. The chatbot retrieves relevant reports, gives simple answers, and can summarize multiple reports in one response. It also provides basic predictions about possible future service disruptions based on past data and trends, helping users stay informed more easily.

4. Community Reporting Feature

The Community Reporting System allows users to submit reports related to infrastructure and environmental conditions in their area. These include power outage reports, water interruption reports, internet outage reports, and weather-related observations. This enables citizens to quickly share information about local issues, helping improve awareness and response to community disruptions.

5. Additional Features

The system includes an intelligent location search bar that allows users to quickly navigate to specific cities and municipalities. This feature provides an intuitive, visual way to explore location-specific information without manually zooming across large map

regions. Typing a known location name is significantly faster than panning and zooming, especially when the target area is far from the current map view (Furnas & Jul, 1997), and text-based location search reduces cognitive load during time-sensitive tasks, allowing operators to go directly to specific cities or municipalities without getting "lost" in the map interface (Nivala & Sarjakoski, 2007). It will also provide Emergency Contact Directory provides quick access to important hotlines and contact details for emergency and public service offices such as the BFP, PNP, hospitals, disaster response units, and utility providers. It serves as a centralized list within the system, helping users quickly find and contact the right agencies during emergencies to improve response time and public safety.

VI. Mockups

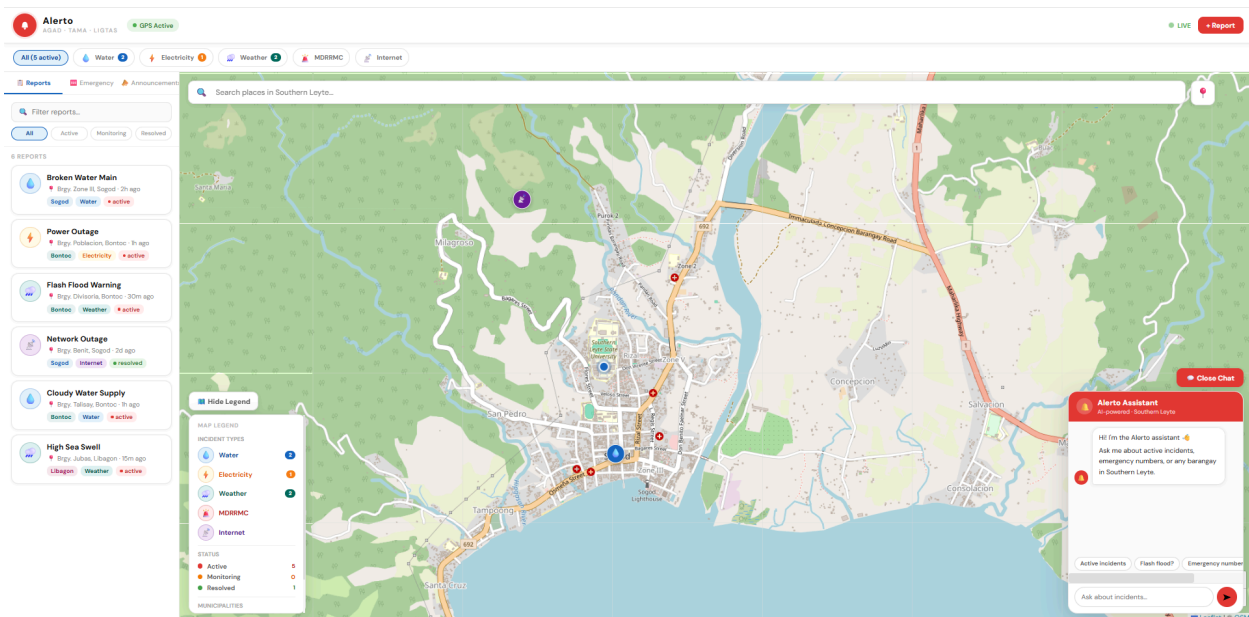


Figure 1.

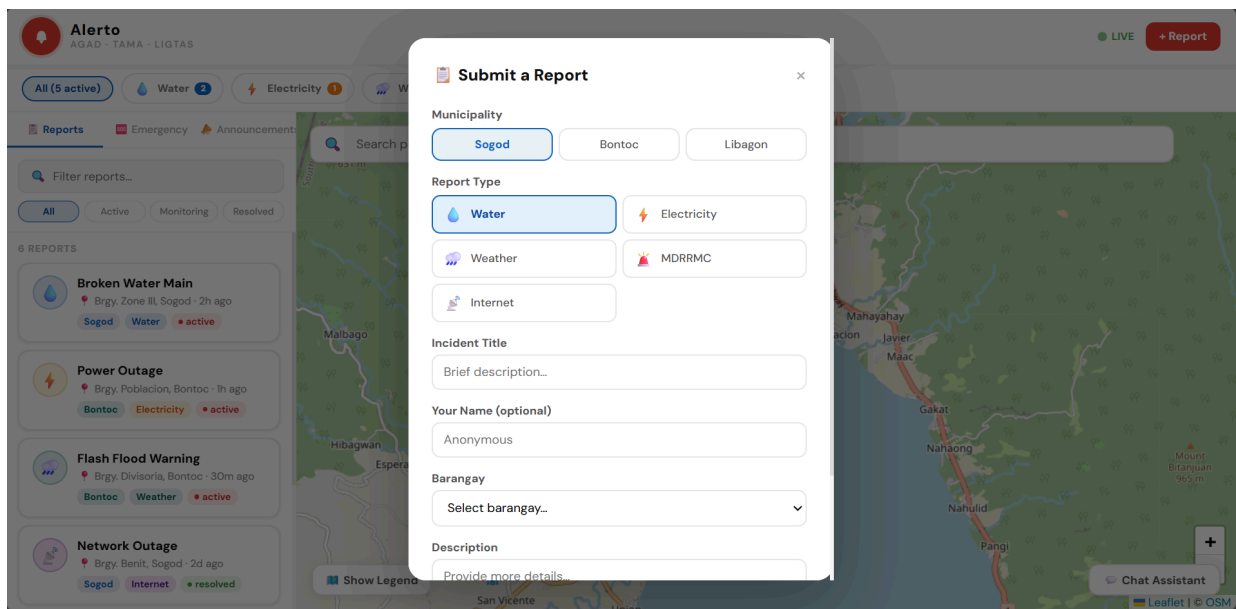


Figure 2.

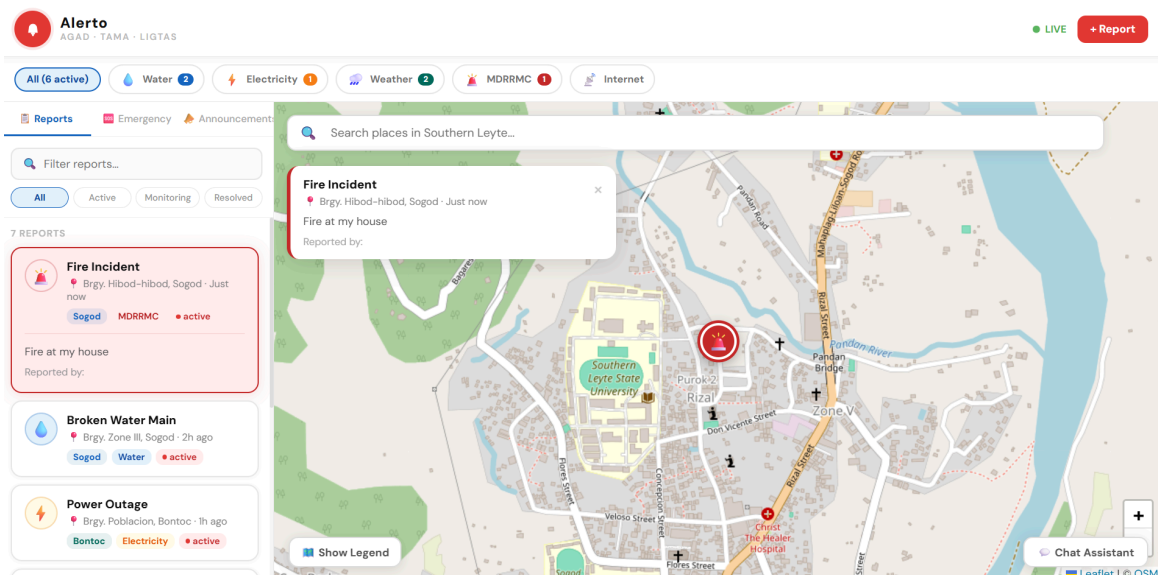


Figure 3.

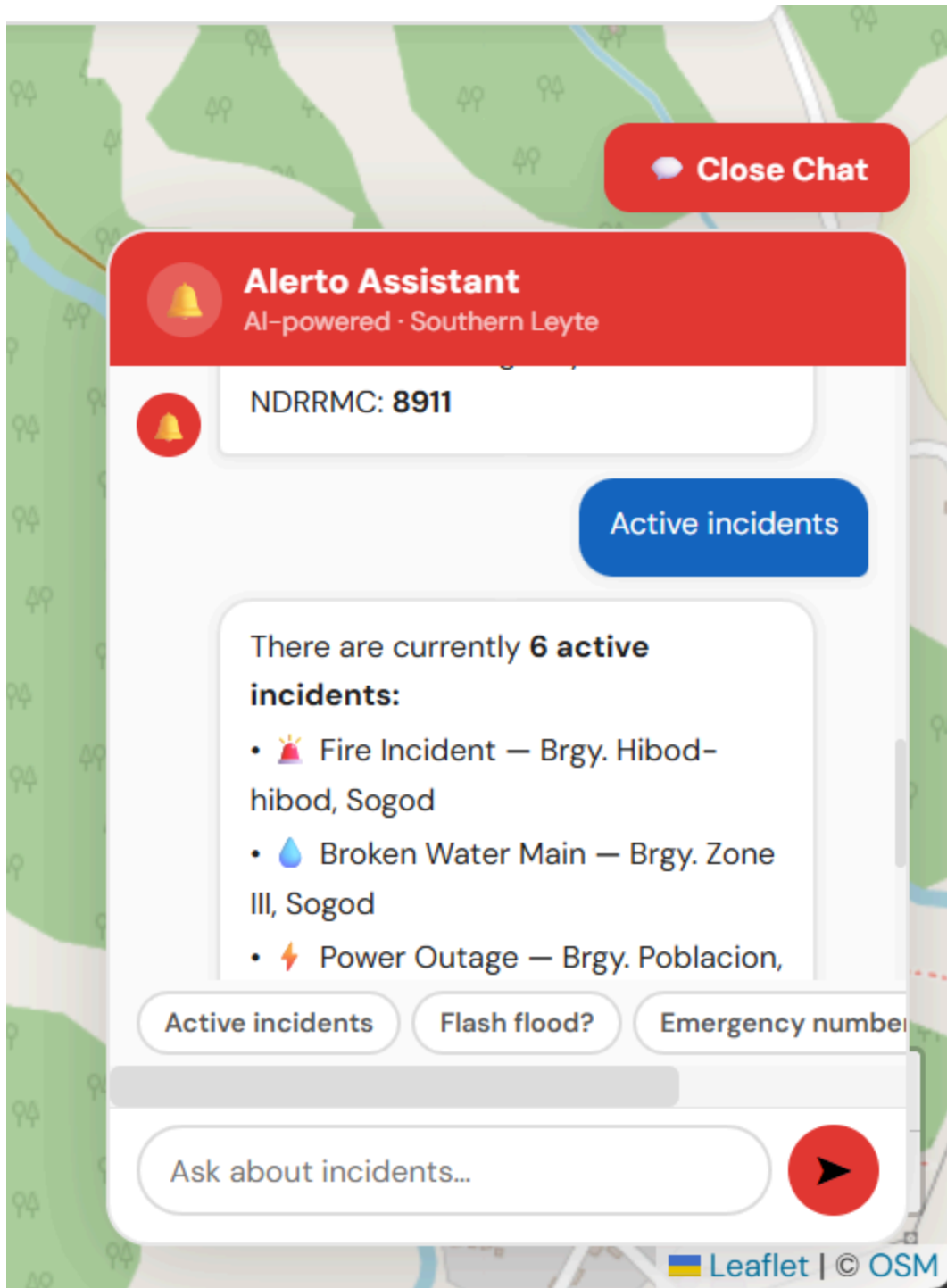


Figure 4.

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